Deliverable 1: Summary of exploratory tasks

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HTML scraping

In task one, we compared logistic principal component regression (PCR) models using just paragraphs and those using both paragraphs and headers (h1–h6). The purpose was to see whether including headers would improve the accuracy of binary class predictions, as headers may contain additional information that is not included in the paragraphs. Both models were tested on the same webpages to make sure the results were comparable. The model that used only paragraphs had an accuracy of 78.1%, meanwhile the model that included headers had an accuracy of 78.3%. This resulted in a 0.233 percentage point improvement, demonstrating that the addition of headers provided a small predictive benefit. However, the difference was minimal, suggesting that the most useful information was in the paragraphs alone.

Bigrams

We tested whether bigrams added predictive information by fitting a second logistic regression using the unigram model's predicted log-odds and 20 bigram principal components. Held-out accuracy decreased from 78.3% to 72.3%. Thus, adding bigram features did not improve classification accuracy in our analysis, suggesting that they offered little additional predictive value beyond individual words in this model.

Neural net

Architecture: The neural network uses a text vectorization layer to convert webpage text into TF-IDF features. The model then applies a dropout layer with a rate of 0.2, followed by a dense hidden layer containing 50 units. A second dropout layer with a rate of 0.2 is applied before the final dense output layer with one unit and a sigmoid activation function. The sigmoid activation produces a probability for binary classification.

Optimization and Loss: The model uses the Adam optimizer to update its weights during training and binary cross-entropy as its loss function. Binary accuracy is used as the evaluation metric during training.

Training Epochs: The model was trained for 5 epochs with a batch size of 32.

Predictive Accuracy: The model achieved a validation accuracy of **80.19%** (0.8018648), correctly classifying approximately 80% of webpages in the validation dataset.